



4.6 The normal distribution

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The normal distribution is a continuous, symmetric, bell-shaped distribution defined by its mean μ and standard deviation σ , written $X \sim N(\mu, \sigma^2)$. Approximately 68% of data lies within 1σ of the mean, 95% within 2σ , and 99.7% within 3σ . Any normal variable can be standardised to the standard normal $Z \sim N(0, 1)$ using $z = (x - \mu) / \sigma$; probabilities are found using a GDC or standard normal tables.

Worked example: follow the reasoning

$X \sim N(50, 8^2)$. Find $P(X < 60)$.
 $z = (60 - 50) / 8 = 1.25$.
 $P(X < 60) = P(Z < 1.25) = 0.8944$ (from GDC/tables).

Quick check · multiple choice

For $X \sim N(100, 15^2)$, which z-value corresponds to $x = 115$?

- A 0.5
- B 1
- C 1.5
- D 2

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The heights of adult men in a population are normally distributed with mean 175 cm and standard deviation 7 cm. [8]
- (a) Find the probability that a randomly selected man is taller than 185 cm.
 - (b) Find the height that is exceeded by only 10% of the population.
 - (c) Find the probability that a randomly selected man has height between 170 cm and 180 cm.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $z = (185 - 175) / 7 = 1.4286$. $P(X > 185) = P(Z > 1.4286) = 1 - 0.9236 = 0.0764$ (≈ 0.0764 , using GDC normal cdf).

(b) Need $P(X > h) = 0.10$, so $P(Z > z) = 0.10$, $z = 1.2816$ (invNorm).

$h = 175 + 1.2816(7) = 175 + 8.97 = 183.97 \approx 184.0$ cm.

(c) $z_1 = (170 - 175) / 7 = -0.7143$, $z_2 = (180 - 175) / 7 = 0.7143$.

$P(170 < X < 180) = P(Z < 0.7143) - P(Z < -0.7143) = 0.7625 - 0.2375 = 0.5250$.

Quick check answer

B. $z = (115 - 100) / 15 = 1$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.